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MohitS704
*Ph.D. Candidate at Johns Hopkins University*
*Last Updated: October 9, 2026*

# Mohit Srivastav

*Curriculum Vitae*

## Education

2022

**Ph.D. Candidate**, Johns Hopkins University (JHU)
Working on the Compact Muon Solenoid (CMS) experiment at the Large Hadron Collider (LHC)
Advisor: Professor Andrei Gritsan

2018–22

**B.S. Physics, B.A. Computer Science**, University of Virginia (UVA)
Highest Distinction in Physics, High Distinction in Computer Science

## Selected Publications

CMS Collaboration. Determination of the spin and parity of all-charm tetraquarks. *Nature*, 648(8092):58–63, 2025.

CMS Collaboration. Measurement of the Higgs boson mass and width using the four-lepton final state in proton-proton collisions at s=13 TeV. *Phys. Rev. D*, 111(9):092014, 2025.

CMS Collaboration. Combined measurements and interpretations of Higgs boson production and decay in proton–proton collisions at $\sqrt{s} = 13\,\mathrm{TeV}$. *Rept. Prog. Phys.*, 89(8):087801, 2026.

CMS Collaboration. Off-shell higgs boson measurements: Yukawa couplings, self-coupling, compositeness, and width. 2026. arXiv preprint https://arxiv.org/abs/2607.23352.

Jeffrey Davis, Andrei V. Gritsan, Lucas S. Mandacaru Guerra, Lucas Kang, Michalis Panagiotou, Jeffrey Roskes, and Mohit Srivastav. Maximizing returns: Optimizing experimental observables at the lhc. 2026. arXiv preprint https://arxiv.org/abs/2601.10822.

Mohit Vir Srivastav, Andrei Gritsan, Nicholas Pinto, Chong Liu, Longsheng Tan, Adinda de Wit, and Li Yuan. Run-2 and Run-3 HZZ K-factors for Experimental Analysis. Technical report, CERN, Geneva, 2024. Work in progress LHCHWG-2026-019.

## Analysis Work

I leverage a variety of techniques from cutting-edge machine learning methods to novel binning techniques to study the properties of particles like the Higgs boson in ways that are sensitive to both quantum interference and spin-parity and perform combined measurements with my collaborators on the CMS experiment.

2023

**Off-shell Higgs Boson**, *CMS Joint 4-Lepton Study Team (CJLST)*

Expert in off-shell Higgs boson analyses on CMS, working towards a full SM EFT interpretation of the off-shell Higgs boson

- Produced world-leading constraints on light-quark Yukawa couplings, the first direct test of Higgs boson compositenessand the Higgs self-coupling via the off-shell region, and experimental discovery of off-shell Higgs boson production [4]
- Collaborated to include an off-shell Higgs analysis into the CMS Higgs Grand Combination for the first time by converting Ref. [2] into the $\kappa$ framework [3]
- Helped produce leading constraints on the width of the Higgs Boson including anomalous terms in gluon-fusion production on CMS [2]
- Craft and utilize machine learning methods, as well as a novel optimal way to merge bins together to ensure sensitivity in multidimensional phasespace [5]

2022–25

**Spin-Parity Analysis of a Tetraquark Candidate**, *CMS B-Physics*

- Found evidence for the spin and parity of 3 low-mass resonances which constitute a candidate for a tetraquark [1]
- Utilized techniques pioneered in the Higgs sector and developed them for use in B-physics

2020–22

**Optimizing Cuts for Experimental Dead-time**, *Mu2e Collaboration*

- Utilized machine learning to minimize experimental deadtime at high beam intensities, culminating in a Thesis in Computer Science
- studied the aging of test-stand di-counters for particle detection

## Technical Work

I help maintain a variety of scientific packages designed for novel work in collider physics and beyond, as well as utilize theorist input to improve Monte Carlo. On the experimental side, I dealt with aligning the tracker during Run-3, and now I assist in efforts to precisely measure assembled modules for the CMS tracker upgrade.

2025

**Lead Developer/Maintainer**, *MiLoMerge Package*

- Developer/maintainer for the MiLoMerge, package, designed to merge bins optimally to reduce separability loss between $N$ arbitrary distributions [5].
- Writing documentation and unit-tests for MiLoMerge in-line with Scikit-HEP requirements
- Version 1.0.0 released on January 15, 2026

2025

**Monte Carlo Simulation Corrections**, *CMS 4-lepton channel*

- Calculated simulation k-factors for the dominant $q\bar{q} \to 4\ell$ background process and $gg \to 4\ell$ Higgs and diboson production processes for the CMS experiment and applied them to real data, as presented in the LHC Higgs Working Group [6]
- Worked in conjunction with experts at Fermilab to establish better parton showering for off-shell Higgs Boson simulation to better match data
- Produced a better filter for the 4-lepton channel to be used in off-shell Higgs Boson simulation

2022

**JHUGen/JHUGen-MELA Development**, *Johns Hopkins University*

- Developer for the JHUGen Package, which includes a Monte Carlo generator as well as a reweighting scheme for experimental high-energy physics, with the latest version (v7.6.0) released Jul. 31, 2026
- Created new Python bindings using PyBind11 to release a version of JHUGen-MELA in Python
- Writing documentation for the JHUGen-MELA package utilizing Doxygen
- Introducing a new set of matrix-element calculations into JHUGen-MELA through the utilization of Madgraph matrix-elements for a range of different processes.

2022

**CMS Tracker Alignment**, *Tracker Alignment Group*

- Utilize survey constraint data to provide initial constraints for the High-Luminosity LHC tracker upgrade alongside teams at Cornell and Fermilab
- Provide alignments for the CMS tracker and validate performance over time as one of the two groups on CMS doing tracker alignment.

## Workshop/Conference Organizing and Professional Development

2026 **Science Policy and Advocacy for Research Competition (SPARC)**
Participant in the SPARC competition organized by the Universities Research Association (URA), to assist "early career scientists to help bridge the gap between science and the public".

2026 **LPC EFT Workshop Local Organizing Committee**
On the local organizing committee for the LPC EFT workshop held at Johns Hopkins.

2024 **JHU Teaching Academy**
Completed the 2024 JHU Teaching Academy, which focused on advancing the development of university-level educators.

## Public Talks & Posters

2026 **From Peak to Tail: Probing the Higgs Boson Width and New Physics with Off-Shell Four-Lepton Production**
Meeting of the Division of Particles and Fields – Fermi National Accelerator Laboratory, IL

2026 **Binning Bonanza: Metric Design for Quantum Observables and Efficient Dimensionality Reduction**
Pheno 2026 – Pittsburgh, PA

2026 **Status of k-factors for Run-3 analyses**
LHC Higgs Cross Section Working Group – LHC Higgs Working Group

2025 **Maximizing Impact: The Quest for Optimal Observables in Experiments**
JHU Experimental Particle Physics Seminar – Johns Hopkins University, MD

2025 **Blasting Off(-shell): $H^* \to ZZ \to 4\ell$ for Precision Higgs boson Measurements at CMS**
Fermilab LHC Physics Center Forum – Fermi National Accelerator Laboratory, IL

2025 **Measurement of the Higgs boson width using the ZZ final state**
Fermilab Annual Users and Affiliates Meeting – Fermi National Accelerator Laboratory, IL

2025 **JHUGen-MELA Tutorial for Higgs SM EFT**
LPC EFT Workshop – Fermi National Accelerator Laboratory, IL

2024 **JHUGen-MELA Tutorial for Higgs SM EFT**
LPC EFT Workshop – University of Notre Dame, IN

2024 **JHUGen-MELA Tutorial for Off-shell Higgs Study**
LPC Off-shell Higgs Boson Workshop – Fermi National Accelerator Laboratory, IL

2023 **Analysis of Hadronic Resonance Structures in the $J/\psi J/\psi$ Invariant Mass Spectrum**
American Physical Society April Meeting – Minneapolis, MN

2022 **Improvements to Cosmic Muon Identification Using Machine Learning**
Virginia Space Grant Consortium Research Symposium – Norfolk, VA

2021 **Improvements to Cosmic Muon Identification Using Machine Learning**
Meeting of the Division of Particles and Fields – Florida State University, FL

## Awards & Honors

2026 **DPF 2026 Student Travel Award (June 2026)**
Conference travel award to go to the American Physics Society meeting of the Division of Particles and Fields 2026 Symposium

2026 **Best overall poster presentation (Jun. 2026)**
Award for best poster presentation at the US CMS meeting at the University of Maryland

2026 **Paul G. Agnew Teaching Award (May 2026)**
Department-level teaching award for continued excellence in teaching

2026 **Krieger School of Arts and Sciences Excellence in Teaching Award (Apr. 2026)**
College-wide teaching award for excellence in teaching by a graduate student

2026 **Phenomenology 2026 Symposium Student Travel Award (April 2026)**
Conference travel award to go to the Phenomenology 2026 Symposium

2025 **LPC Guest & Visitor Award (May 2025)**
Awarded monetary support to be at Fermilab to conduct central shifts at the LHC Physics Center (LPC) and assist with metrology for the outer tracker upgrade

2025 **E.J. Rhee Flair in Teaching Award (May 2025)**
Department teaching award for "creative flair" in teaching

2025 **Kupperman Family Department Travel Grant (May 2025)**
Department award to facilitate travel to summer conferences

2025 **Universities Research Association Visiting Scholars Program (Mar. 2025)**
Universities Research Association (URA) award to stay at Fermilab for June/July 2025 to perform research into off-shell EFT, conduct shifts, and metrology for the tracker upgrade

2024 **CMS IRIS-HEP Analysis Software Training Travel Grant (Mar. 2024)**
Awarded monetary support to travel to Princeton university to attend the IRIS-HEP software training as well as the US CMS meeting

2023 **URA P5 Fermilab Travel Grant (Dec. 2023)**
Awarded monetary support to travel to the P5 policy plan announcement town hall at Fermilab

2021 **Mitchell Undergraduate Research Scholarship (Jun. 2021)**
Awarded a scholarship to conduct research at the University of Virginia under Professor E. Craig Dukes for the Mu2e experiment

2021 **Virginia Space Grant Consortium Research Scholarship (Apr. 2021)**
Awarded a scholarship for conducting research into improving dead-time for the Mu2e experiment

2021 **SULI Argonne Lab (Jan. 2021)**
Offer extended; did not accept

2020 **DAAD Rise Research Fellowship (Mar. 2020)**
Offer extended; cancelled due to COVID-19

2020 **SULI Jefferson Lab (Jan. 2020)**
Offer extended; cancelled due to COVID-19

## Selected Teaching Experience

**Head TA**, *Classical Mechanics for Physics Majors*, Fall 2024, 2025, 2026
Head TA for Physics 1 for Physics Majors

**TA**, *General Physics Lab*, Fall 2023 – Spring 2024
TA for General Physics Lab 1 & 2

**TA**, *Computer Algorithms*, Summer 2020 – Spring 2021
TA for CS 4102 (Computer Algorithms) at the University of Virginia

**TA/Head TA**, *Introductory Computer Science in Python*, Spring 2019 – Spring 2022
TA and Head TA for CS 1110/1111 (intro to computer science) at the University of Virginia

## Physics Outreach

2026

**Fairfax Science and Engineering Fair Category Judge (March 2026)**

Judging high school students at the regional science fair in Fairfax, Virginia

2025 – 26

**Volunteer**, *Johns Hopkins Physics Fair*, April 2026, 2026

Performed demos such as cloud chambers and "ping-pong cannons" while talking about various physics concepts for local families.

2019 – 22

**Volunteer**, *UVA National Physics Day show*, December 2019, April 2022

Performed demos while talking about various physics concepts for local families every year the fair happened at UVA.

## Other Experience

2023–24

**Fitness Monitor**, *Ralph O'Connor Rec Center*, JHU

Helped manage facilities, play music, and interact/help patrons at the gym

2022–23

**SAT Tutor**, *The Answer Class*, Maryland

SAT Tutor leading prep courses in various classes at high schools in the Baltimore area

2020–22

**Assistant**, *UVA Physics Department Facilities*, UVA

Assisted the department lecture demonstrator with daily tasks such as locking the building, moving demos, and some administrative tasks.

2020–22

**Writer**, *The Yellow Journal*, UVA

Collaboratively wrote articles and headlines for the Yellow Journal, the University of Virginia's satirical publication

2019–21

**Treasurer**, *International Relations Organization*, UVA

Managed taxes and finances on the order of $30k for the 501c3 International Relations Organization at the University of Virginia